



Egyptian Academy for Engineering & Advanced Technology
Mechatronics Engineering Department



**“Design and Simulation of a 3-DOF Fire Detection and
Tracking Robotic Arm”**



Course Name

Robotics (1) - MEC 354

Design of Mechatronic Systems - MEC 353

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1. Abstract

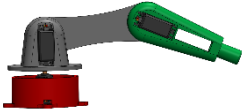
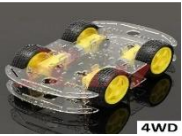
This project presents the design and simulation of a 3-DOF robotic arm for fire detection and tracking applications using MATLAB Simulink and Simscape Multibody. The system utilizes flame sensors to detect fire locations and generate a corresponding base rotation offset for directional alignment. A predefined circular trajectory is defined in Cartesian space and transformed into joint space using inverse kinematics, while forward kinematics is applied for position validation of the end-effector.

The control architecture is implemented in a simulation environment, where the robotic arm responds to different sensor activation scenarios, including single and multiple fire detections. The integration of sensor-based decision making with kinematic modeling enables accurate fire tracking and smooth robotic motion. Simulation results demonstrate stable performance and reliable tracking behavior, confirming the effectiveness of the proposed system for autonomous fire response applications.

Keywords

3-DOF robotic arm, Fire detection system, Fire tracking, MATLAB Simulink, Simscape Multibody, Inverse kinematics, Forward kinematics, Sensor-based control, Autonomous robot, Robotic manipulation.

2. Components

Name	Number	Image
Arduino uno	1	
Flame Sensor	3	
3DOF Arm Robot	1	
Water Pump (5V) And Water Tank	1	
Motor Driver l298n	1	
BMS	1	
Gear Motors	2	
Car Chassis	1	
3.7v Batteries	3	
Ultrasonic Sensor	1	
MG996R Servo Motor	3	

3. Introduction

Robotic systems are widely used in hazardous environments where human intervention may be dangerous or inefficient. One of the most important applications is fire detection and response, where fast and accurate targeting of the fire source can significantly reduce damage and improve safety. This project presents the design and simulation of a 3-DOF robotic arm mounted on a mobile platform for fire detection and tracking applications.

The system uses flame sensors placed at different angular positions to estimate the direction of the fire source. Based on these sensor readings, a control algorithm generates a base rotation offset to orient the robotic arm toward the detected fire location. The arm motion is defined using a circular trajectory in Cartesian space and converted into joint space using inverse kinematics. Forward kinematics is then used to validate the end-effector position and ensure model consistency.

The robotic arm is mounted on a mobile cart to improve flexibility and extend its operational workspace. The entire system is implemented and tested using MATLAB Simulink and Simscape Multibody, providing a realistic simulation environment for both kinematic modeling and control design. This setup allows evaluation of the system under different fire detection scenarios, including single and multiple sensor activations.

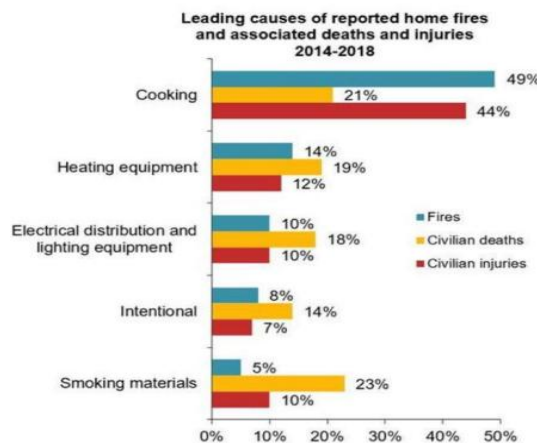


Figure 1 Causes of Fire Accidents

4. Objectives

The main objective of this project is to design and simulate a 3-DOF robotic arm mounted on a mobile platform for fire detection and tracking applications. The system aims to utilize flame sensors to detect the direction of fire sources and generate an appropriate base rotation offset to align the robotic arm toward the detected location. Another objective is to develop a reliable control strategy that translates sensor information into accurate robotic motion.

The project also focuses on implementing inverse kinematics to compute the required joint angles from a predefined Cartesian trajectory, while forward kinematics is used to verify the accuracy of the end-effector position. In addition, a smooth circular trajectory is designed to ensure stable motion of the robotic arm during operation. Finally, the complete system is implemented and tested using MATLAB Simulink and Simscape Multibody to evaluate its performance under different fire detection scenarios, including single and multiple sensor activations.

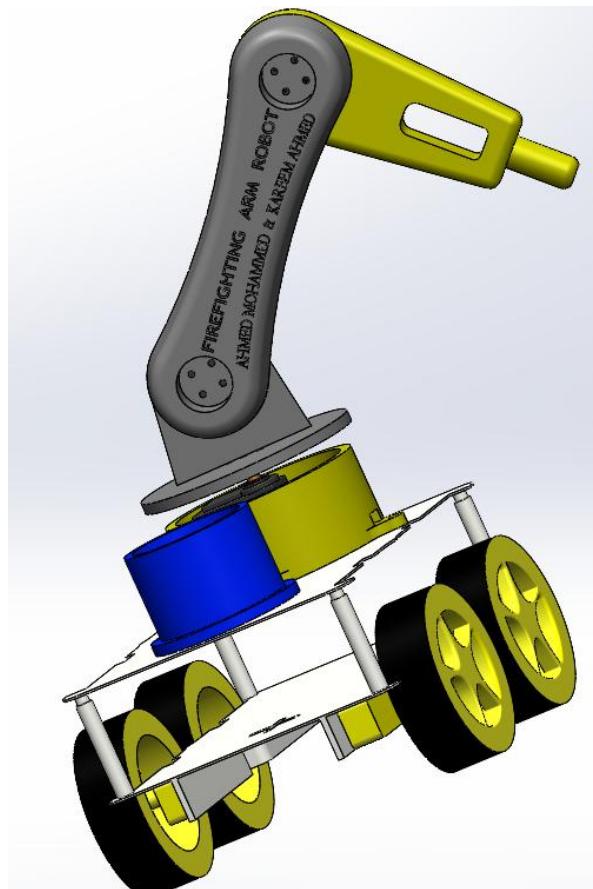


Figure 2 Firefighting Robot

5. System Overview

The proposed system consists of a 3-DOF robotic arm mounted on a mobile platform designed for fire detection and tracking applications. The overall architecture integrates sensing, decision-making, kinematic modeling, and control within a unified simulation environment using MATLAB Simulink and Simscape Multibody.

The sensing layer is based on flame sensors placed at different angular positions around the robot to detect the presence and approximate direction of fire sources. The outputs of these sensors are processed by a Fire Controller, which determines whether the system should be enabled and computes a base rotation offset to orient the robotic arm toward the detected fire direction.

The motion planning layer generates a predefined circular trajectory in Cartesian space for the end-effector. This trajectory is designed to ensure smooth and continuous motion during operation. The trajectory is then converted into joint space using inverse kinematics to compute the required joint angles for the 3-DOF arm.

The control layer ensures accurate execution of the desired motion, while forward kinematics is used to validate the actual position of the end-effector and confirm system consistency. The entire model is implemented and tested in Simscape Multibody, which provides a realistic physical simulation of the robotic structure and its dynamic behavior.

6. Software Tools and Technologies

Simulink blocks such as PID controllers, switches, and enable subsystems are used to implement the control logic and ensure stable system performance. The combination of these tools provides a complete environment for designing, testing, and validating the robotic fire detection and tracking system.

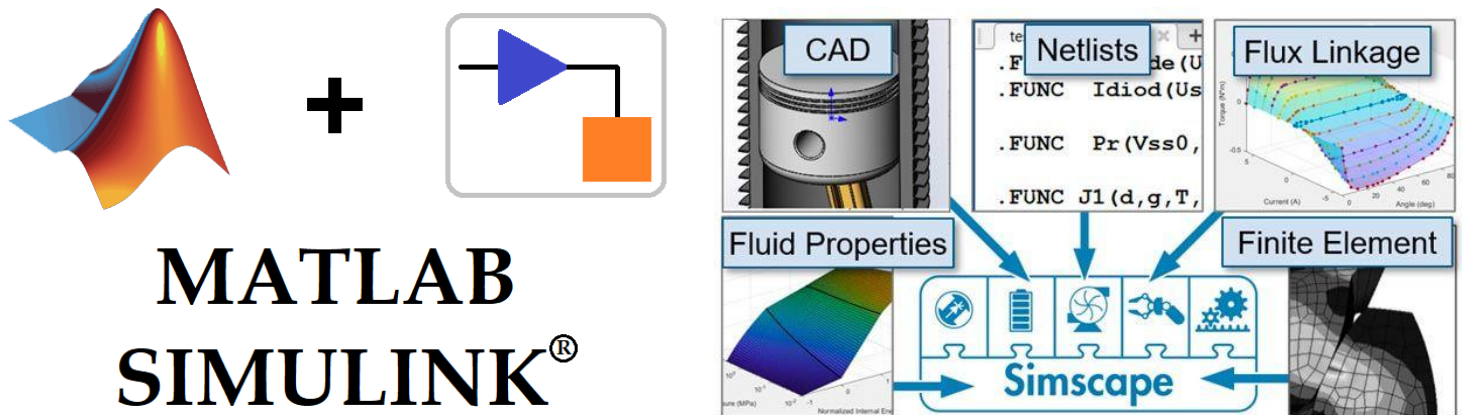


Figure 3 Simulation Programs

7. Hardware Implementation

The hardware part is one of the crucial parts in the development of firefighting robot. It includes Arduino UNO, IR flame sensors, servo motors, submersible water pump, motor driver, mini breadboard, and rubber wheels. Fig 4 shows the block diagram of firefighting robot which consists of three IR flame sensors as the input of the system. Arduino UNO is used as a micro-controller that connects other components. L298n Motor driver is used to drive motors and is capable of running two DC motors (Left DC motor and Right DC motor) at the same time.

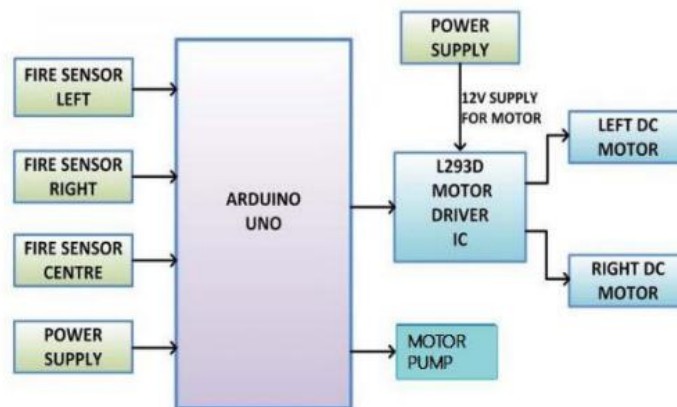


Figure 4 Block Diagram of Firefighting Car

7.1 HARDWARE USED

Fig 5 shows the Arduino UNO board. It is basically a micro-controller kit that is used to get data from peripheral devices (sensors, motors, etc.). The Arduino UNO Micro-controller board is based on the ATmega328P IC. The ATmega328P is good platform for robotics application which makes robot to extinguish fire in real time. Arduino UNO board consist the sets of digital and analog pins that may act as an interface to various expansion boards and other circuits. It contains everything needed to support the microcontroller.



Figure 5 Arduino UNO

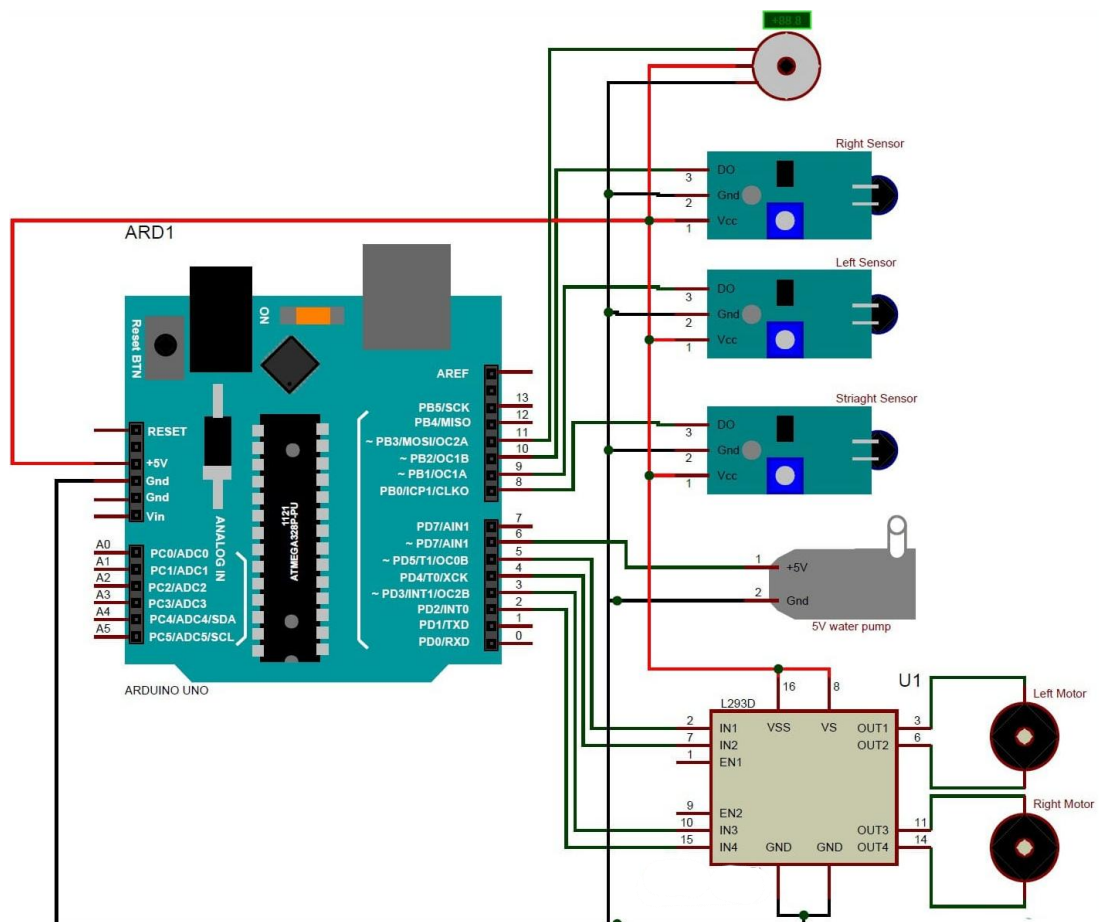


Figure 6 Digital car Circuit

Fig 7 shows the IR Flame Sensor. The IR flame sensor senses the environment and detects the presence of fire or flame. The module is based on the IR receiver and basically detects the presence of flammable and harmful gases like nitrogen, hydrogen, carbon mono oxide. The signal detection capacity is adjustable. The robot contains three flame sensors.

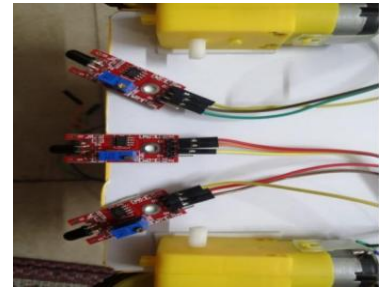


Figure 7 IR Flame Sensor

Fig 8 shows the Servo Motors. Servo Motors are electronic devices that are mainly used for providing specific velocity and acceleration.



Figure 8 servo motors

Fig 9 shows the Submersible Water Pump. Submersible Water Pump is ideal for making automatic watering system using Arduino. The water pump is an important part of the robot as it will pump water to extinguish the fire.



Figure 9 Submersible Water Pump

Fig 10 shows the BO Motor. BO Motor is a dual shaft motor having 300rpm. It converts electrical energy into mechanical energy. It is the replacement to our metal gear DC motors. Our robot uses four dual shaft motors.



Figure 10 BO Motor

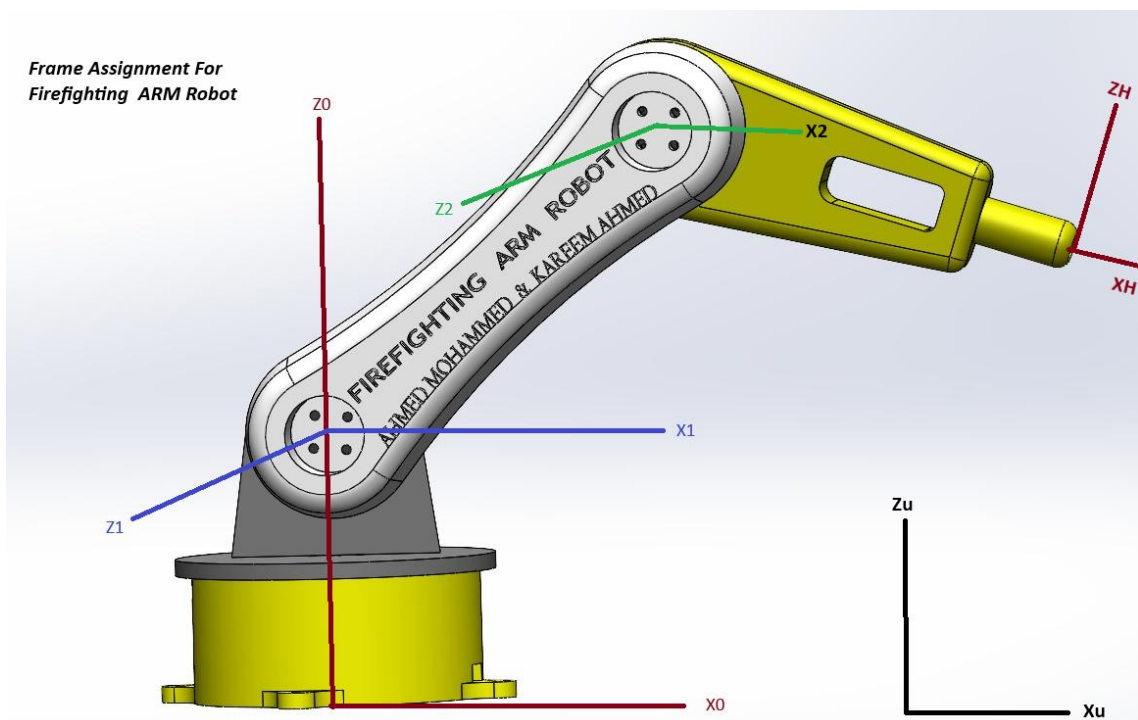
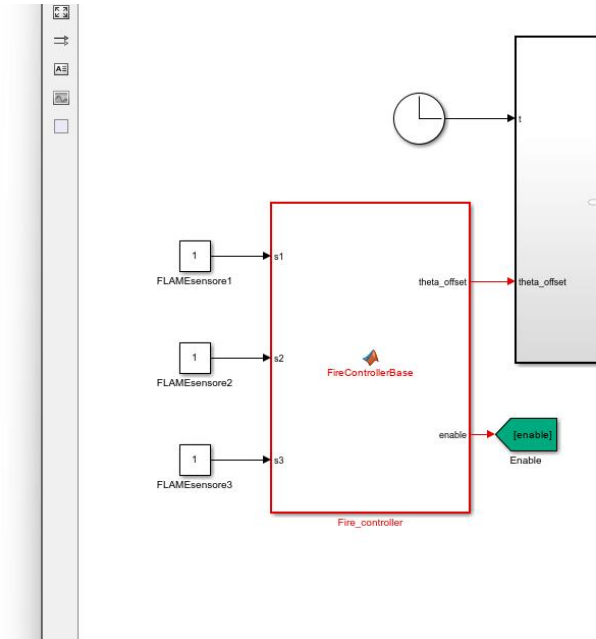


Figure 11 3DOF Robot Arm

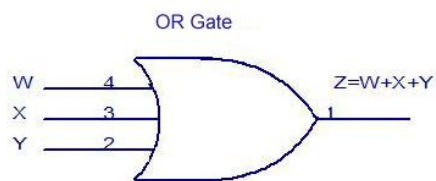
Fig 11 shows the 3DOF Robot Arm. The 3DOF robotic arm is a mechanical manipulator with three degrees of freedom that allows movement in multiple directions. It is driven by three servo motors, which control the base rotation, arm lifting, and end-effector positioning. The robotic arm is mounted on the robot chassis and is used to accurately direct the water pump toward the fire source. By adjusting the angles of the servo motors, the arm can position the nozzle to improve the effectiveness of the firefighting process. This mechanism enables the robot to reach and target the flame more precisely during operation.

8. Simulation





3 Input OR Gate



TRUTH TABLE

INPUTS			OUTPUT
W	X	Y	Z
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Figure 12 The Truth table for the Flame sensores

9. Code Explanation

9.1 Fire Controller Base Function

```
function [theta_offset, enable] = FireControllerBase(s1, s2, s3)

% Fire Controller for Base Rotation

% Inputs:
%   s1, s2, s3  -> Flame sensors (0 = no fire, 1 = fire)

% Outputs:
%   theta_offset -> Desired base rotation angle (rad)
%   enable -> 1 = robot (ON), 0 = robot (OFF)

% Sensor angular positions around the robot (in radians)
sensor_angles = [0, 90, 180] * pi/180;

% Default State
enable = 1; % assume system is active

% No Fire Detected -
if (s1 == 0 && s2 == 0 && s3 == 0)
    theta_offset = 0; % no direction
    enable = 0; % disable robot
    return;
end

% Sensor Activation Vector
sensor_states = [s1 s2 s3];

% Safety Check
if sum(sensor_states) == 0
    theta_offset = 0;
    enable = 0;
else
    |
    % Weighted Average Direction
    % Computes average direction based on active sensors

    theta_offset = sum(sensor_states .* sensor_angles) / sum(sensor_states);
end
end
```

9.2 Desired Trajectory 3DOF Function

```
function [xd, yd, zd] = Desired_Trajectory_3DOF(t)

% Robot base height
d1 = 110.55;

% Motion period
T = 1.5;

% Circle radius
R = 60;

% Vertical oscillation amplitude
h = 0;

% Circle center position
xc = 160;
yc = 0;

% Base height offset for trajectory
zc = d1 + 60;

% Angular velocity
w = 2*pi/T;

% X trajectory (circular motion in XY plane)
xd = xc + R*cos(w*t);

% Y trajectory (circular motion in XY plane)
yd = yc + R*sin(w*t);

% Z trajectory (constant or small sinusoidal motion)
zd = zc + h*sin(w*t);
|
end
```

9.3 Inverse Kinematics 3DOF Function

```
|function [theta1_d, theta2_d, theta3_d] = Inverse_Kinematics_3DOF(xd, yd, zd, theta_offset)

% Robot link parameters
d1 = 110.55;
a1 = 194;
a2 = 160;

% Base rotation with sensor offset
theta1_d = atan2(yd, xd) + theta_offset;

% Planar distance to target point
r = sqrt(xd^2 + yd^2);

% Vertical distance relative to base height
z = zd - d1;

% Cosine law term for elbow angle
D = (r^2 + z^2 - a1^2 - a2^2) / (2 * a1 * a2);

% Clamp value to avoid numerical errors
D = max(min(D, 1), -1);

% Elbow configuration (stable solution)
theta3_d = atan2(-sqrt(1 - D^2), D) + deg2rad(140);

% Shoulder angle computation
theta2_d = atan2(z, r) - atan2(a2 * sin(theta3_d), a1 + a2 * cos(theta3_d)) + pi;

end
```

9.4 Forward Kinematics 3DOF Function

```
|function [x_a, y_a, z_a] = Forward_Kinematics_3DOF(theta1, theta2, theta3)

% Robot link parameters
d1 = 110.55;
a1 = 194;
a2 = 160;

% Projection of the arm on the XY plane
r = a1*cos(theta2) + a2*cos(theta2 + theta3);

% End-effector position in XY plane
x_a = r * cos(theta1);
y_a = r * sin(theta1);

% End-effector position in Z direction
z_a = d1 + a1*sin(theta2) + a2*sin(theta2 + theta3);

end
```

10. Conclusion

This project demonstrated the successful design and simulation of a 3-DOF fire detection and tracking robotic arm mounted on a mobile platform. By integrating flame sensors with a sensor-based control strategy, the system effectively determines fire direction and aligns the robotic arm in real time. The use of inverse and forward kinematics ensures accurate motion execution, while the implementation in MATLAB Simulink and Simscape Multibody provides realistic and reliable system validation.

The results confirm that the proposed system achieves stable, smooth, and responsive performance under different detection scenarios, highlighting its effectiveness as a foundation for autonomous fire response applications.